

Scheduled Task Threat Detection and Mitigation in Azure using Microsoft Sentinel SIEM

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146640

A Computer Networks and Cyber Security Project II submitted to the School of Computing and Engineering Sciences in partial fulfilment of the requirements for the award of the Bachelor's Degree in Computer Networks and Cyber Security of Strathmore University

School of Computing and Engineering Sciences

Strathmore University

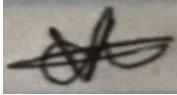
Nairobi, Kenya

July 2024

Declaration and Approval

I declare that this work has not been previously submitted and approved for the award of a bachelor's degree by this or any other University. To the best of my knowledge and belief, the project contains no material previously published or written by another person except where due reference is made in the project itself.

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Acknowledgement

I would like to extend my heartfelt thanks and deep appreciation to the individuals who have offered me invaluable support and guidance throughout the course of this project.

I express my gratitude to Mr. Humphrey Owuor for his exceptional guidance, mentorship, and expertise. His feedback, encouragement, and belief in my abilities have driven the success of the project. His dedication to my research growth and development has been inspiring, and I am grateful for the invaluable lessons learned under his supervision.

I express my gratitude to my family for their unwavering love, understanding, and support throughout my journey. Their belief in me and sacrifices have been a constant source of motivation. Their encouragement during challenging and triumphant moments has been indispensable, and I am forever grateful for their presence in my life. I am grateful to all these individuals for their invaluable contributions, encouragement, and belief in my abilities. Their support has been instrumental in the successful completion of this project.

Abstract

This project implemented Security Information and Event Management (SIEM) capabilities in cloud environments using Microsoft Sentinel, with a specific focus on detecting and mitigating threats associated with scheduled tasks. As businesses increasingly rely on Microsoft Azure for critical services and infrastructure, the risk posed by malicious actors exploiting scheduled tasks to execute persistent threats becomes a significant concern. This project created a scheduled task that mimics an attack, which created a baseline for distinguishing between benign and malicious scheduled tasks and implementing an automated mitigation strategy using Azure Sentinel. By making use of Sentinel's analytics and automated response features, the project sought to improve threat detection accuracy, minimize false positives, and ensure the prompt isolation of compromised virtual machines (VMs). This solution aimed to provide a framework for improving cloud security in Azure environments, thereby protecting organizations from potential data breaches, financial losses, and reputational damage.

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List of Abbreviations

SIEM	Security Information and Event Management
VM	Virtual Machine
SOAR	Security Orchestration Automated Response
APTs	Advanced Persistent Threats
SSAD	Structured Systems Analysis and Design Method
KQL	Kusto Query Language
NSG	Network Security Groups
AAD	Azure Active Directory
LAW	Log Analytics Workspace
RDP	Remote Desktop Protocol

Chapter 1: Introduction

1.1 Background Information

Microsoft Azure is a widely used cloud computing platform that offers a range of services, including virtual machines, databases, storage, and networking. Its security features make it a preferred choice for businesses of all sizes, enabling them to deploy and manage enterprise-level IT infrastructure efficiently (Verma et al., 2019). However, the widespread adoption of Azure also makes it a significant target for cyber threats, necessitating proper security measures to protect its infrastructure.

This project focused on scheduled tasks which are a fundamental feature of operating systems allowing users and administrators to automate routine operations such as backups, updates, and maintenance tasks (Kannan et al., 2017). In Windows environments, the Task Scheduler service facilitates the creation, modification, and execution of these tasks based on predefined triggers and schedules. While scheduled tasks improve system efficiency and reliability, they also present a security risk. Malicious actors can exploit scheduled tasks to perform unauthorized actions, persist on a compromised system, escalate privileges, or execute malicious code at specific times (Jang-Jaccard, J., & Nepal, S. 2018). Detecting and mitigating these threats is crucial to maintaining the integrity and security of IT environments.

Microsoft Sentinel is a cloud native SIEM (Security Information and Event Management) and SOAR (Security Orchestration Automated Response) solution that offers toolsets for implementing security in cloud environments. SIEM systems are essential components of modern cybersecurity frameworks, providing real-time analysis of security alerts generated by applications and network hardware. They compile data from multiple sources, offering a centralized view of an organization's security posture. Microsoft Sentinel can be used to identify potential security incidents and reduce false positives. Sentinel's threat detection capabilities and automated response actions enable organizations to set up SIEM systems to mitigate risks, ensure a more secure cloud environment (Tuyishime, E et al., 2023).

By implementing Sentinel's analytics and response actions, organizations can better detect and mitigate threats posed by malicious scheduled tasks within Azure environments. This integration improves the overall security posture, ensuring that potential threats are identified and addressed promptly, thereby minimizing the risk of widespread damage. This project aimed to implement SIEM functionality in cloud environments by using Microsoft Sentinel to

specifically target and manage the threats associated with scheduled tasks, providing a more secure cloud infrastructure

1.2 Problem Statement

The problem this project addressed is the manipulation of scheduled tasks within Microsoft Azure environments by malicious actors. These tasks, designed for routine operations, can be exploited to execute APTs (Advanced Persistent Threats), posing a significant security threat (Uçar, Mesut. 2023). According to studies, attackers can use scheduled tasks to persist in the system, elevate privileges, and spread malware, often going undetected due to their legitimate appearance.

Ideally, scheduled tasks should only be used for benign purposes, such as maintenance, backups and updates. However, the reality is that attackers can and do exploit these tasks to carry out harmful activities. This creates a critical gap between the intended use of scheduled tasks and their potential misuse. The consequences of this gap are severe, leading to compromised systems, data breaches, and significant financial and reputational damage to affected organizations. Businesses relying on cloud, for example Azure, for their operations are particularly at risk, as a single compromised VM can threaten the entire network's security (Z. S. Zainudin et al 2018).

1.3 Aim

The aim of this project was to implement the detection and mitigation of malicious scheduled tasks within Microsoft Azure environments using tools found in Microsoft Sentinel, thereby improving overall cloud security.

1.4 Specific Objectives

- i. To investigate the ability of Microsoft Sentinel's Security Information and Event Management (SIEM) in detecting and mitigating threats posed by scheduled tasks in cloud environments.
- ii. To assess the challenges associated with detecting malicious scheduled tasks in Azure and review existing solutions.
- iii. To design and develop an Azure Sentinel solution to distinguish between benign and malicious tasks by implementing tools components in Azure Sentinel.
- iv. To test the developed solution to determine its ability to identify and isolate malicious scheduled tasks.

1.5 Research Questions

- i. How well does Microsoft Sentinel work in detecting and mitigating threats from scheduled tasks compared to traditional SIEM systems in cloud environments?
- ii. What challenges exist in detecting malicious scheduled tasks in Azure environments?
- iii. How can Azure Sentinel be utilized to differentiate between benign and malicious scheduled tasks?
- iv. How well does the developed solution work in real-world scenarios for identifying and mitigating malicious scheduled tasks?

1.6 Justification

This project was crucial because it addressed a security gap in azure cloud environments that are widely used across various industries. By implementing the detection and mitigation of malicious scheduled tasks, this project aimed to protect organizations from potentially devastating cyberattacks. The benefits of this project extended to improving the overall security posture of businesses and institutions, ensuring the integrity of their operations, and protecting sensitive data.

Moreover, this project contributed to the field of cybersecurity by providing a practical solution to an unacknowledged but hazardous problem. The development and testing of an automated detection and response system within Azure Sentinel, as done in this project, serves as a blueprint for similar initiatives, offering valuable insights and methodologies that can be replicated and adapted across different contexts. Ultimately, this project aimed to create a more secure cloud environment, reducing the risk of cyber threats and improving the resilience of cloud-based infrastructures.

1.7 Scope and Limitations

1.7.1 Scope

This project implemented the detection and mitigation of malicious scheduled tasks specifically within Microsoft Azure environments. The scope included the assessment of current challenges, design and development of a solution using Azure Sentinel, and the testing of the implemented solution. The project covered azure cloud platform and scheduled tasks attacks in azure only and this focused approach ensured a thorough investigation and solution development.

1.7.2 Limitations

The limitations included variability in the types of attacks executed using scheduled tasks and the evolving nature of cyber threats, which may affect the solution's long-term effectiveness. Additionally, the research relied on the capabilities of Azure Sentinel and the accuracy of its analytics, which posed constraints. However, these limitations were acknowledged, and the project provides a good foundation that can be adapted and improved as needed to address emerging threats.

Chapter 2: Literature Review

2.1 Introduction

The literature review chapter covers existing research and technologies to enhance Security Information and Event Management (SIEM) capabilities in cloud using Microsoft Sentinel specifically for detecting threats and mitigating attacks from scheduled tasks. This review is organized based on the objectives of the study, starting with an overview of the area of study, including what is SIEM, what is Microsoft Sentinel, and the risks of scheduled tasks in cloud. The chapter then goes into the specific problems the study wants to address, how these affect the organization and what has been done so far to mitigate them. Finally, the review investigates existing solutions and studies in the field and identifies the gaps that the proposed solution addresses.

2.2 Microsoft Sentinel SIEM

SIEM systems are a key part of modern cybersecurity architecture by providing real time analysis of security alerts from applications and network devices. SIEM systems aggregate and correlate data from multiple sources to give a single view of an organisation's security posture. This is key to identifying patterns and anomalies that may indicate security incidents so you can respond quicker and manage threats better (Kavanagh & Rochford, 2020).

In the context of cloud computing Microsoft Azure is a leading platform that provides virtual machines, databases, storage and networking. Azure's scalability, flexibility and security features make it a popular choice for businesses. But with Azure being so widely used it's also a bigger target for cyber attacks so there is a need to have advanced security to protect the infrastructure (Diogenes, et al., 2022).

Microsoft Sentinel is a cloud-native Security Information and Event Management (SIEM) solution designed to provide security analytics and threat intelligence across an enterprise. It can integrate with various Microsoft services and third-party solutions to offer visibility and protection. Sentinel can be used in data collection by integrating with numerous sources, including azure services, on-premises systems, and other cloud environments. It can be configured to use Azure Log Analytics as the backend for storing and querying log data, improving its data handling capabilities.

Additionally, regarding threat detection, Microsoft Sentinel offers a platform for analytics using custom rules written in Kusto Query Language (KQL) to help identify threats. Users can customize detection rules and queries to tailor the system to their specific needs. In this project,

sentinel this feature is used to detect potential security incidents caused by scheduled tasks and can be used to reduce false positives (Copeland, M., 2021).

In terms of incident response, Sentinel supports automated actions through connecting to playbooks created using Azure Logic Apps, this can enable quick containment and mitigation of threats. Investigation tools such as workbooks, notebooks, and a timeline view help in analysing incidents and tracing attack paths, while integrated case management features assist in collaboration among security teams.

Sentinel threat intelligence abilities come from being able to incorporate feeds from various sources and creating alerts with additional information to prioritize and understand threats better. Its cloud-native architecture leverages Azure's infrastructure for scalability, high availability, and performance, with elastic scaling capabilities to handle large data volumes and high query loads.

All these features combined can be used in effectively detecting and mitigating malicious scheduled tasks. Scheduled tasks in windows environments, controlled by the task scheduler service, allow users and administrators to automate things like backups, updates and maintenance tasks. While these tasks make the system more efficient and reliable, they are a big security risk if an attacker can use them. (Sharma, M et al., 2022).

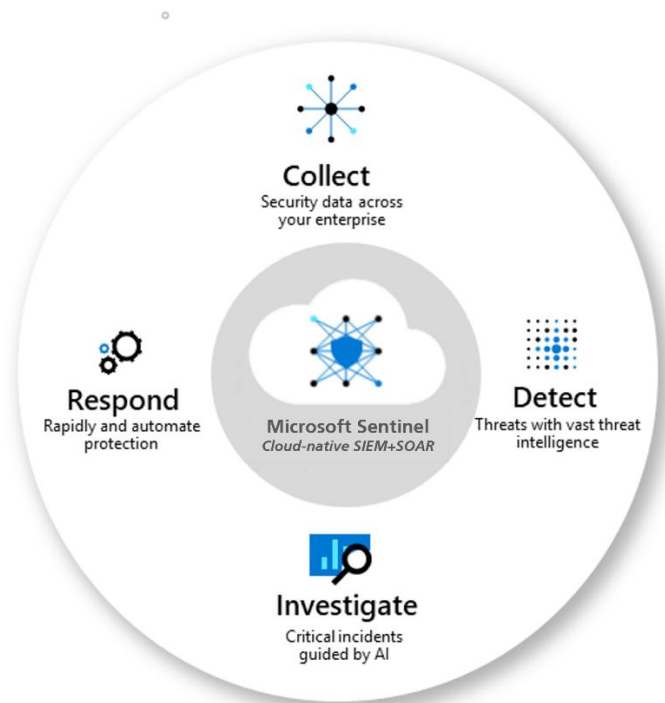


Figure 2.1 Microsoft Sentinel

2.3 Scheduled Tasks Threat Detection

This study addressed the problem of malicious actors exploiting scheduled tasks to perform unauthorized actions in cloud environments. These threats can compromise the integrity and security of resources in the cloud, resulting in data breaches, financial losses and reputational damage. Scheduled tasks are exploited because they are trusted processes in the operating system and because of this, malicious actors often take advantage of these legitimate scheduling tools and techniques, making it difficult to distinguish between normal administrative tasks and malicious activities.

Additionally, without an established baseline of normal scheduled tasks and their behaviour, it can be difficult to identify anomalies. Scheduled tasks may interact with other systems and services within the azure environment, and this affects businesses by introducing vulnerabilities that can be exploited for extended period without detection. If not addressed, the consequences are continuous threats in the system, increased attack surface and potential widespread damage in case of an attack (Zainudin & Abdul Molok, 2018).

While Azure provides several security features and tools, they may not cover all aspects of scheduled task management. Previous attempts to solve this problem are developing threat detection algorithms and monitoring systems. However, those solutions lack accuracy in differentiating between malicious and benign scheduled tasks. Furthermore, although there is improvement in detecting suspicious activity, there is still a gap in automating response to mitigate the different identified threats (Lee et al., 2023).

2.4 Existing Solutions

To address the challenges of detecting and mitigating threats related to scheduled tasks in cloud environments existing Security Information and Event Management (SIEM) solutions were reviewed. This section reviews three SIEM systems: Splunk Enterprise Security, IBM QRadar and ArcSight ESM. These solutions are analysed based on their capabilities, technology used and effectiveness in addressing malicious scheduled tasks issues in cloud environments.

2.4.1 Splunk Enterprise Security

Splunk Enterprise Security (ES) is a well known SIEM solution with strong data analytics capabilities. Splunk ES provides real-time monitoring, advanced threat detection and incident response by aggregating and analysing data from multiple sources including logs, network devices and endpoints. Splunk's machine learning algorithms identify patterns and anomalies that indicate potential security threats, it's a powerful tool for security professionals (Ananthapadmanabhan, A., 2023).

Splunk ES integrates with cloud services including Microsoft Azure, AWS and Google Cloud Platform in cloud environments. This allows organizations to monitor and secure their cloud infrastructure. However, Splunk ES is strong in data analytics but can be complex to manage and requires a lot of resources for deployment and maintenance. Even with its capabilities there may still be challenges in differentiating between benign and malicious scheduled tasks as there is no established base of established baseline of normal scheduled tasks and their behaviour, and malicious tasks. (Ananthapadmanabhan, A., 2023).

2.4.2 IBM QRadar

IBM QRadar is another SIEM solution known for its capabilities in threat detection, incident response, and compliance management. It provides visibility into network traffic, user activities, and endpoint behaviour, helping organizations identify and respond to security threats effectively. QRadar's strengths include its analytics, machine learning algorithms, and ability to correlate vast amounts of data to detect complex attack patterns. (Barhami, Y. Et al., 2023).

However, despite its strengths, IBM QRadar faces challenges in scaling and integrating with specialized cloud services, particularly for automating mitigating strategies for detected malicious scheduled tasks. In cloud environments, the dynamic and distributed nature of resources necessitates seamless integration and scalability, areas where QRadar can encounter difficulties. The system's deployment and resource management complexities can hinder its effectiveness in real-time threat detection and response, especially for unexpected threats like malicious scheduled tasks.

2.4.3 ArcSight ESM

ArcSight Enterprise Security Manager (ESM) by Micro Focus is a SIEM solution that focuses on centralised security monitoring and advanced threat detection. ArcSight ESM aggregates and correlates security data from multiple sources and provides real-time analysis and response to security incidents. Its data analytics and machine learning capabilities can detect sophisticated threats.

ArcSight ESM supports deployment in cloud environments including Microsoft Azure and AWS. This allows organizations to extend their security monitoring and threat detection to their cloud infrastructure. However, there is challenge in finding a single service that can specifically be used for detection and mitigation of malicious scheduled tasks in azure.

Additionally, no clear strategies have been set dealing with detecting threats associated with scheduled tasks (Sekharan, S., et al 2022).

2.5 Gaps

There are notable inadequacies in addressing the specific challenge of dealing with persistent attacks.

Table 2.1 Gaps

Other Works	Cited article	Gaps Identified	Proposed solution
The paper "Advanced Persistent Threats (APT): evolution, anatomy, attribution and countermeasures" studies different types of persistent attacks.	Sharma, A., Gupta, B.B., Singh, A.K. et al. (2023).	This paper does not specify scheduled tasks as a way used by attackers to execute malicious tasks on a machine.	The use of malicious scheduled task to achieve persistence in the machines and its significant consequences were well researched, documented and tested.
The paper "Towards the Automatic and Schedule-Aware Alerting of Internetwork Time Series," discusses monitoring and alerting for attacks.	D. Perdices, J. L. García-Dorado, J. Ramos. et al. (2023).	This paper focuses on monitoring and alerting of administrators in the event of an attack. It does not focus on providing a mitigation solution that can minimise the effect of the attack	This project used Sentinel and integrated it with with Azure Logic App to implement seamless automation to detect and isolate a compromised

			VM to mitigate malicious scheduled tasks, a type of persistent attack
The paper "Cloud Storage Defense Against Advanced Persistent Threats: A Prospect Theoretic Study," This paper discusses ways to defend against persistent attacks in the cloud.	L. Xiao, D. Xu, C. Xie, N. B. Mandayam and H. V. Poor, (2017)	This paper uses prospect theory (PT) is applied to formulate the interaction between the defender of a cloud storage system and an attacker. This method leaves room for various attacks as it only defends against attacks that can be anticipated or prospective attacks	This project used log analytics workspace to connect to virtual machines thus analysing all tasks executed to detect any kind of attack

2.6 Conceptual Framework

The SIEM solution created involved integrating Microsoft Sentinel with azure resources to improve on its SIEM capabilities, specifically targeting the detection and mitigation of threats associated with scheduled tasks. The framework involves the following process:

First is scheduled task execution, where a scheduled task is executed on the Azure VM, generating a security event log. This is followed by log collection, the logs from the VM are collected and sent to azure log analytics workspace. Next is log analysis in azure sentinel, the collected logs are analysed using custom analytic rules and custom scripts to identify suspicious behaviour. A decision is then made, if the task is benign the system returns to log collected to determine if the other log events are benign or malicious, if no task was executed the system waits for tasks to be executed on the VM and if task is malicious the system moves to the fourth step. The fourth step is detection and alert generation, when Sentinel identifies a malicious task, its configured to generate an alert that triggers the response. The fifth step is the automated

response: The alert triggers an automated response configured using the azure logic app tool which isolates the affected VM to mitigate the threat.

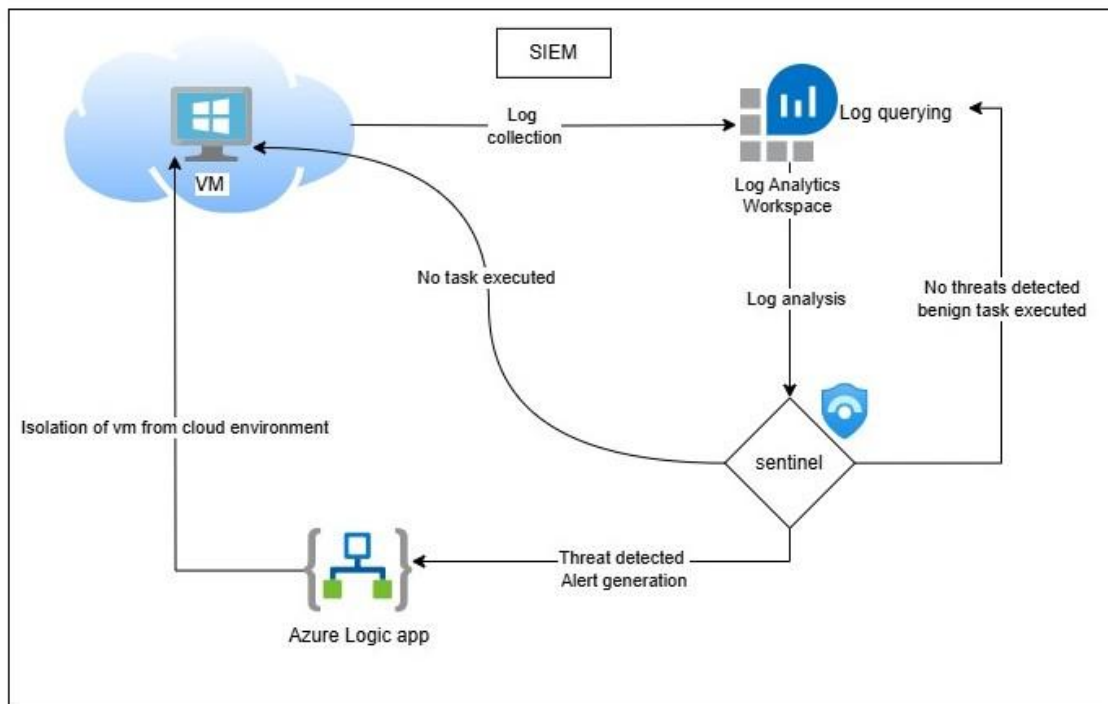


Figure 2.2 Conceptual Framework

Chapter 3: Research Methodology

3.1 Introduction

This chapter details the methodology used for implementing SIEM capabilities in the cloud using Microsoft Sentinel for detecting and mitigating threats from scheduled tasks. It used the Structured Systems Analysis and Design Method (SSAD) to provide a systematic approach to the analysis and design of the solution. The Agile methodology was chosen because of its flexibility and iterative nature, allowing for continuous feedback and adaptation throughout the project. The chapter covers the phases of ideation and requirement analysis, design, development, and testing, leading to the delivery of a well implemented and functional SIEM.

3.2 Research Paradigm

The research paradigm chosen for this project was the Structured Systems Analysis and Design Method (SSAD). SSAD is a methodological approach that involves breaking down complex systems into smaller, manageable components through a series of systematic phases. It includes detailed analysis and design stages, ensuring a thorough understanding of the system requirements and a structured path to system development (Dennis, A., Wixom, B., & Tegarden, D. 2015).

SSAD was particularly suited for this project as first, its structured approach allowed accurate documentation of the requirements and processes involved in implementing SIEM capabilities with Microsoft Sentinel. For instance, the detailed examination of scheduled task threats requires a clear and systematic method to identify potential vulnerabilities and define how these threats can be detected and mitigated. SSAD allowed for this by providing tools and techniques to map out the processes, data flows, and system interactions. Its systematic approach ensured thorough documentation, precise analysis, and detailed design, making it the best choice for this project.

3.3 Methodology

The methodology chosen for this project was the Modified Waterfall methodology. The Modified Waterfall approach was selected because it combines the structured, phase-based progression of the traditional Waterfall model with the flexibility of iterative feedback, making it well-suited for complex systems like SIEM that require planning and adaptability to evolving threats and requirements.

The Modified Waterfall methodology follows a sequential design process but allows for iterations between phases, enabling adjustments based on testing and feedback. This approach

ensures thorough documentation and clear progression while accommodating necessary changes. In this project it was used to develop the system incrementally, focusing on core functionalities first logging, then detection of malicious scheduled tasks, then alerting, and finally isolation. This methodology allowed the system processes to be followed as clear steps are followed in the development process of this methodology. The modified waterfall methodology provides a feedback system, there the project can move back to the previous step, if there are any errors.

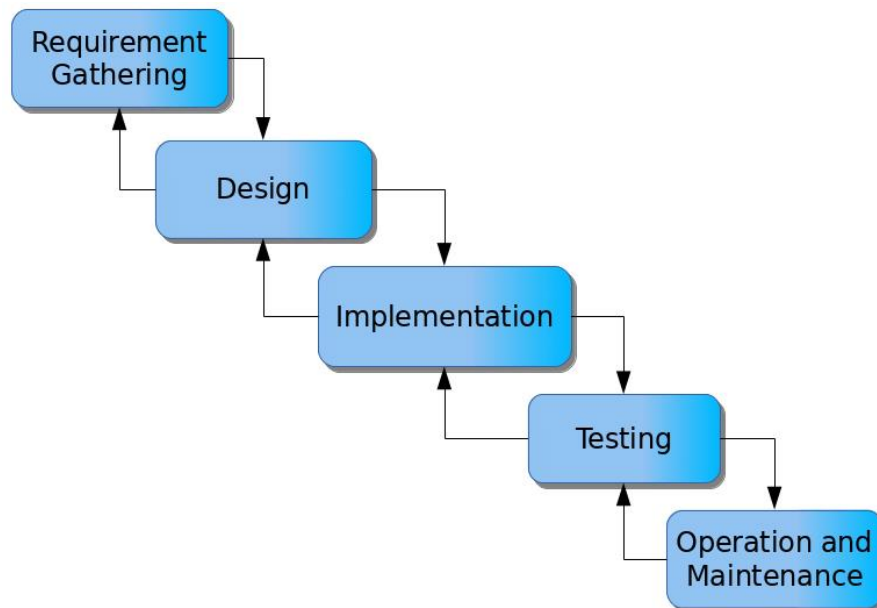


Figure 3.1 Modified Waterfall Methodology

3.3.1 Requirement Gathering

This phase was important for understanding the problem and defining the necessary functionalities and constraints of the system to be implemented. It involved gathering information on Security Information and Event Management (SIEM), scheduled task threats, and cloud security best practices. It also involved identifying the data required to implement an SIEM that successfully detects and mitigates attacks launched using scheduled tasks.

SIEM systems aggregate and analyse log data from various sources within an IT infrastructure to detect and respond to potential security threats. Important features include real-time monitoring, threat detection through correlation and pattern recognition, incident response automation, and reporting for compliance purposes. Knowing how to integrate the SIEM with the existing environment, its scalability, and its capabilities for threat detection and response was important.

Data collection began with the analysis of logs generated by the execution of a scheduled task on the target virtual machine. These logs included security events from virtual machines (VMs), and network security group (NSG) flow logs. The security events from VMs capture detailed information about scheduled task executions, when they were created, the user that created them, what task is executed and its respective path. These logs are ingested into Microsoft Sentinel where they are analysed to identify patterns that suggest malicious activity. The data acquired from these logs is crucial in designing and implementing a solution that can be used to identify malicious scheduled attacks and create a mitigation strategy (Zannone, N et al., 2023).

3.3.2 Design

The research approach utilized Structured Systems Analysis and Design Method (SSAD), which supports the implementation of the SIEM system. SSAD is a method of analysis and design of a system that models it as a group of interacting systems, each broken down into manageable components through a series of systematic phases (Saad, A et al., 2019). This approach was particularly beneficial for interconnected systems like the one in this project as it allows for clear representation of system functionalities and interactions, making the system easier to manage, extend, and modify. This method is particularly suited for this project's SIEM system as it is made up of different components coming together to create a whole solution as seen the system architecture discussed below.

SSAD was employed to systematically address the design of the SIEM system using Microsoft Sentinel for scheduled task threat detection in a cloud environment. The system's architecture was also visualized and documented using various analysis diagrams discussed in Chapter 4

System Architecture

The architecture of the SIEM using Microsoft Sentinel was designed to leverage cloud-native services for log ingestion, analysis, and threat response. The primary components and their interactions are as follows:

- i. **Log Collection:** Various Azure services, including VMs, Network Security Groups (NSGs), and Azure Active Directory (AAD), generate logs that capture security events, network traffic, and user activities. These logs are collected and sent to Microsoft Sentinel.

- ii. **Log Ingestion and Storage:** Microsoft Sentinel ingests these logs into a centralized Log Analytics workspace. This workspace serves as a repository for all collected log data, allowing for efficient querying and analysis.
- iii. **Data Analysis and Threat Detection:** Sentinel uses configured analytics rules to process and analyse the ingested log data. These rules, written in Kusto Query Language (KQL), are designed to identify suspicious activities, such as unusual scheduled task creations or modifications.
- iv. **Alert Generation:** When an analytics rule detects a potential threat, Sentinel triggers an alert that is configured. This alert contains detailed information about the suspicious activity and triggers a response mechanism.
- v. **Automated Response:** The alert activates a playbook implemented using Azure Logic Apps. The playbook automates response actions, such as notifying security personnel, enriching alert data, and initiating containment procedures.
- vi. **VM Isolation:** As part of the containment process, the playbook isolates the compromised VM by deactivating its network interface card (NIC), applying restrictive NSG rules, and removing the VM from any load balancers. This action effectively contains the threat and prevents further malicious activities.

Using SSAD and cloud-native tools, the system was designed to be scalable, maintainable and efficient in detecting and mitigating scheduled task threats within a cloud environment.

3.3.3 Tools and Techniques

Key tools and technologies used include:

- i. **Microsoft Sentinel:** This was ideal because of its cloud-native SIEM capabilities, Sentinel offers the ability to implement threat detection and automated response mechanisms, and seamless integration with other Azure services. Its ability to analyse large volumes of data in real-time made it an ideal tool for this project.
- ii. **Azure Logic Apps:** This was configured to automate responses to detected threats. Logic Apps allow the creation of automated workflows that can trigger the isolation of compromised virtual machines, notify security personnel, and perform other remediation actions without manual intervention.
- iii. **Azure Monitor:** This tool was used for collecting, analysing, and acting on data from the cloud and on-premises environments. Azure Monitor helps in tracking the performance and health of the solution, ensuring that it operates efficiently.

- iv. **Kusto Query Language (KQL):** KQL was used within Sentinel to create custom analytics rules and queries for detecting suspicious scheduled task activities. Its powerful querying capabilities enabled detailed analysis of log data to identify patterns indicative of security threats.

3.4 Testing

To confirm that the solution is fit for purpose, a testing strategy was used to ensure the system can detect and mitigate threats originating from scheduled tasks. The testing process involved two types of tests, each designed to validate different aspects of the system.

- i. **Integration Testing:** This involved testing the interaction between different system components to ensure they work seamlessly together. The integration between the virtual machine, Microsoft Sentinel, and Azure Logic Apps was tested to verify that logs are correctly ingested, analysed, and trigger appropriate responses. Integration testing ensures that the system's components interact correctly and that data flows smoothly between them. The specific type of integration testing done included:
 - Executing tasks to show that logs are being collected using log analytics workspace and displayed and analysed in sentinel.
 - Executing a malicious task to show that the alert generated using sentinel triggers a response in azure logic apps.
- ii. **Scenario Testing:** To test whether the system worked as intended, both benign and malicious scheduled tasks were executed. These included tasks designed to mimic common attack patterns, such as persistence mechanisms and data exfiltration attempts. The system was observed to determine whether it can accurately detect and differentiate between benign and malicious activities. Additionally, whether the compromised virtual machine (VM) was correctly isolated as part of the automated response was tested. The specific scenarios tested included:
 - Executing a benign task to determine whether it is detected as benign and a malicious task to determine whether it is detected as malicious.
 - Executing a malicious task to determine whether the response action is performed that is isolation of the compromised virtual machine

3.5 Deliverables

The deliverables of this project were to:

- i. Establish a secure connection between the VMs and Microsoft Sentinel Configure VMs to generate and send security events, network traffic, and user activity logs to Microsoft Sentinel
- ii. Create and configure a centralized Log Analytics workspace in Microsoft Sentinel to store and manage the ingested log data.
- iii. Develop and configure analytics rules using Kusto Query Language (KQL) in Microsoft Sentinel to identify suspicious activities and differentiate between malicious and benign scheduled tasks.
- iv. Set up alerts in Microsoft Sentinel that trigger when analytics rules detect potential threats.
- v. Create and configure a playbook using Azure Logic Apps to automate response actions.
- vi. Test the system by simulating the triggering event, execution of a malicious scheduled task detected as malicious.

Chapter 4: System Analysis and Design

4.1 Introduction

This chapter explains in detail the analysis and design of the SIEM system focusing on the requirements, architecture and diagrams that represent the system's structure, how the system is designed to function, how different components interact, and how data flows through the system. The diagrams include system architecture diagram, network diagram, use case diagram, sequence diagram and flowchart diagram. Functional and non-functional requirements are outlined, providing a clear understanding of the expected capabilities and performance standards of the system.

4.2 System Requirements

4.2.1 Functional Requirements

The functional requirements are the operations and behaviours that the system must support. These include:

- i. **Logging and Reporting:** The system must log all detected events related to scheduled tasks and generate reports for further analysis. This includes detailed logs of task executions, detections and any mitigation actions taken.
- ii. **Detection of Malicious Scheduled Tasks:** The system should be able to detect and identify malicious scheduled tasks within the azure environment. This involves monitoring and analysing logs generated by scheduled tasks and differentiating between benign and malicious scheduled tasks to determine any suspicious activity.
- iii. **Automated Response and Mitigation:** On detecting a malicious scheduled tasks the system should automatically trigger a response, which is isolating the compromised virtual machine to prevent further damage.
- iv. **Integration with Microsoft Sentinel:** The virtual machines and azure logic apps should integrate with Microsoft Sentinel to take advantage of its SIEM capabilities including real-time monitoring, threat detection and response strategies.

4.2.2 Non-Functional Requirements

The non-functional requirements are the quality attributes and performance criteria the system must meet. These include:

- i. **Performance:** The system should provide real time monitoring and analysis of scheduled tasks with minimal latency, ensuring timely detection and response.
- ii. **Accuracy:** The detection algorithm should have a high accuracy rate to minimize false positives and false negatives in identifying malicious tasks.
- iii. **Reliability:** The system should be reliable with minimal downtime and should be able to handle large volumes of log data without failure.
- iv. **Flexibility:** The system should be flexible enough to adapt to changes in the cloud environment such as new types of scheduled tasks.
- v. **Continuous Monitoring:** The system should ensure continuous monitoring for early threat detection and faster incident response.

4.3 System Architecture Diagram

The system architecture diagram provides an overview of how various components of the system interact and work together. It includes the Azure environment, the primary cloud infrastructure where the VMs with scheduled tasks and other resources were hosted. Microsoft Sentinel which integrates with the VMs, Log analytic workspace and Azure logic app for real-time threat detection, logging and automated response. Custom scripts written in sentinel that were used to differentiate between malicious and benign tasks and trigger an alert in case of a malicious task. Finally, Azure logic app which contains a custom playbook that triggered mitigation actions like isolating a compromised VM when an alert of a malicious task is made.

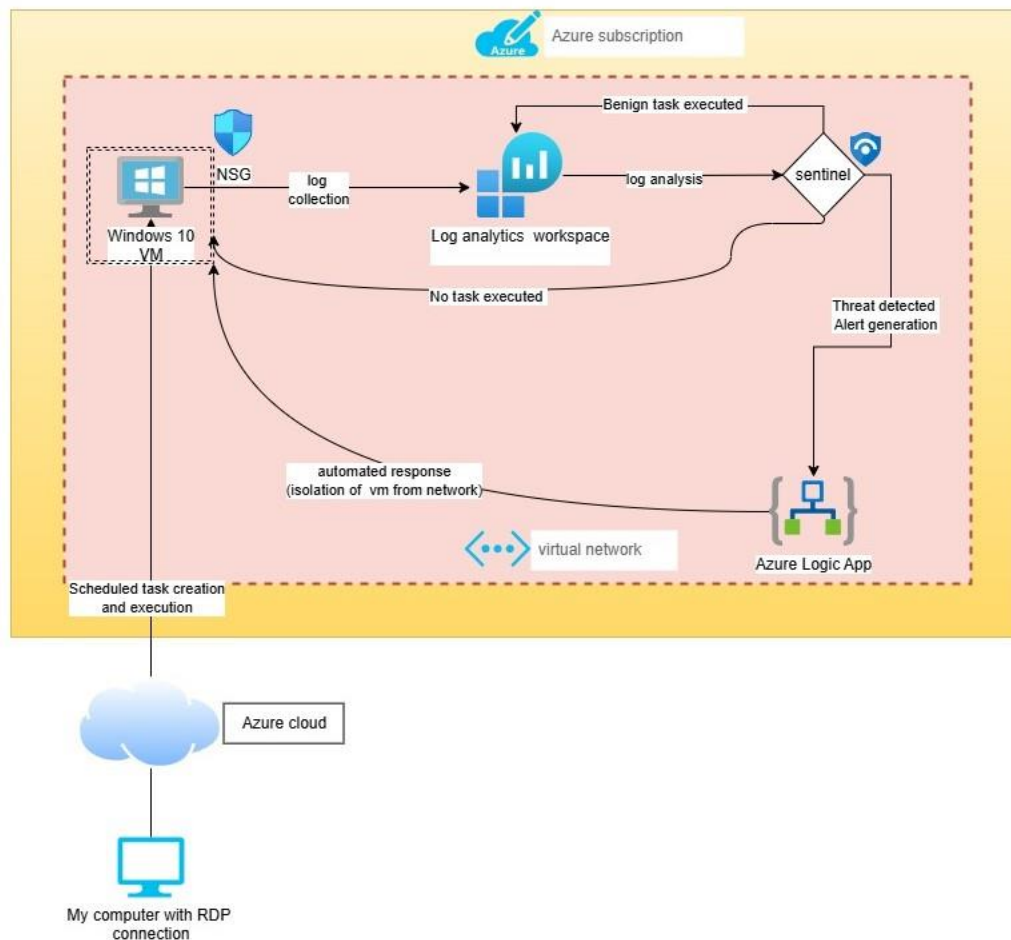


Figure 4.1 System architecture diagram

4.4 System Analysis Diagrams

4.4.1 Network Diagram

Figure 4.2 below shows the network topology of how the project was set up on Azure. When the virtual machine was deployed in Azure, that virtual machine was placed on a Virtual Network (Vnet) and assigned an IP address on that network as well as a network interface. An NSG (Network Security Group) was used to filter network traffic to and from Azure resources.

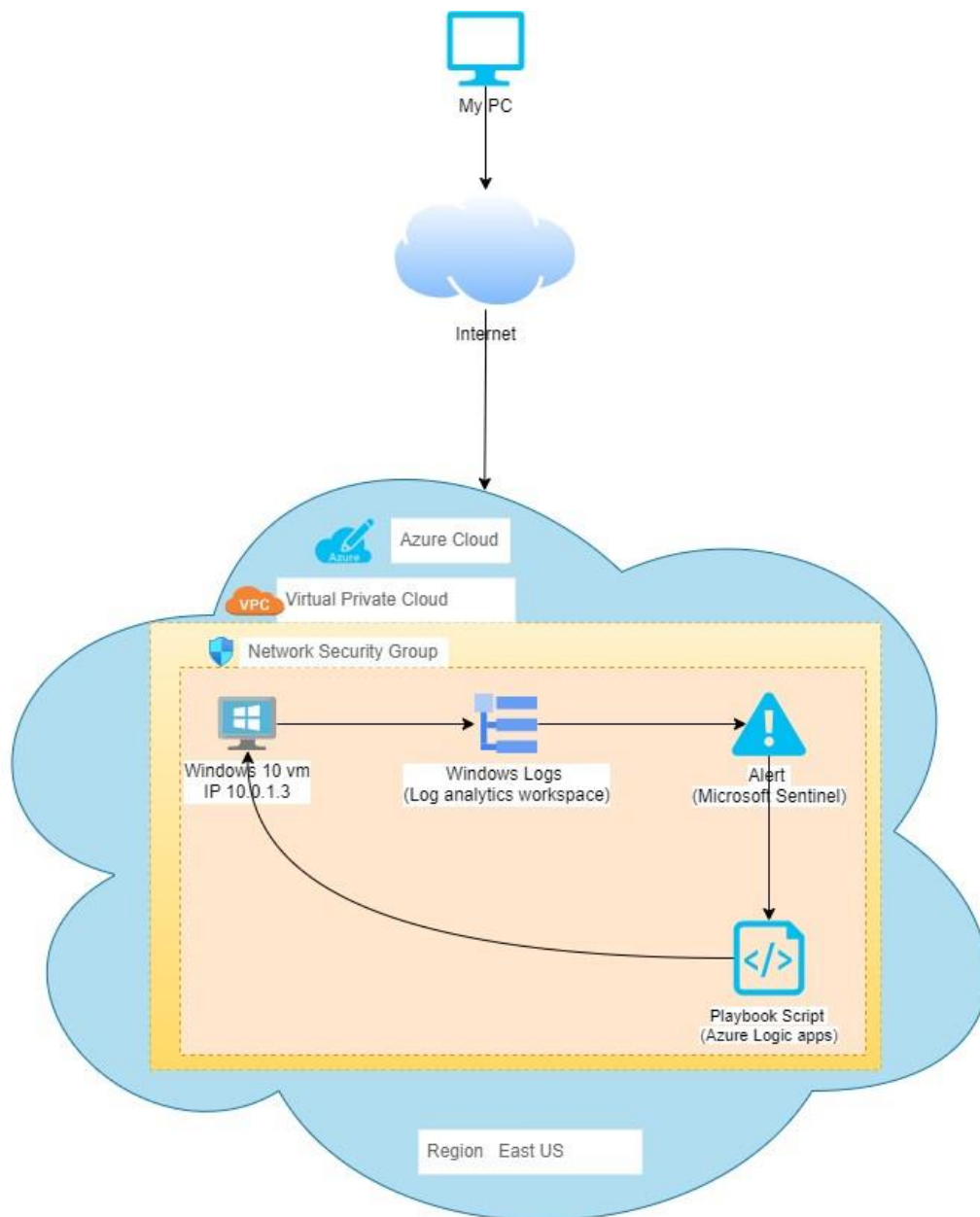


Figure 4.2 Network Diagram

4.4.2 Use Case Diagram

Figure 4.3 represents the interaction between users, system administrators and the system. Users schedule tasks while administrators configure the logging of scheduled tasks in the environment.

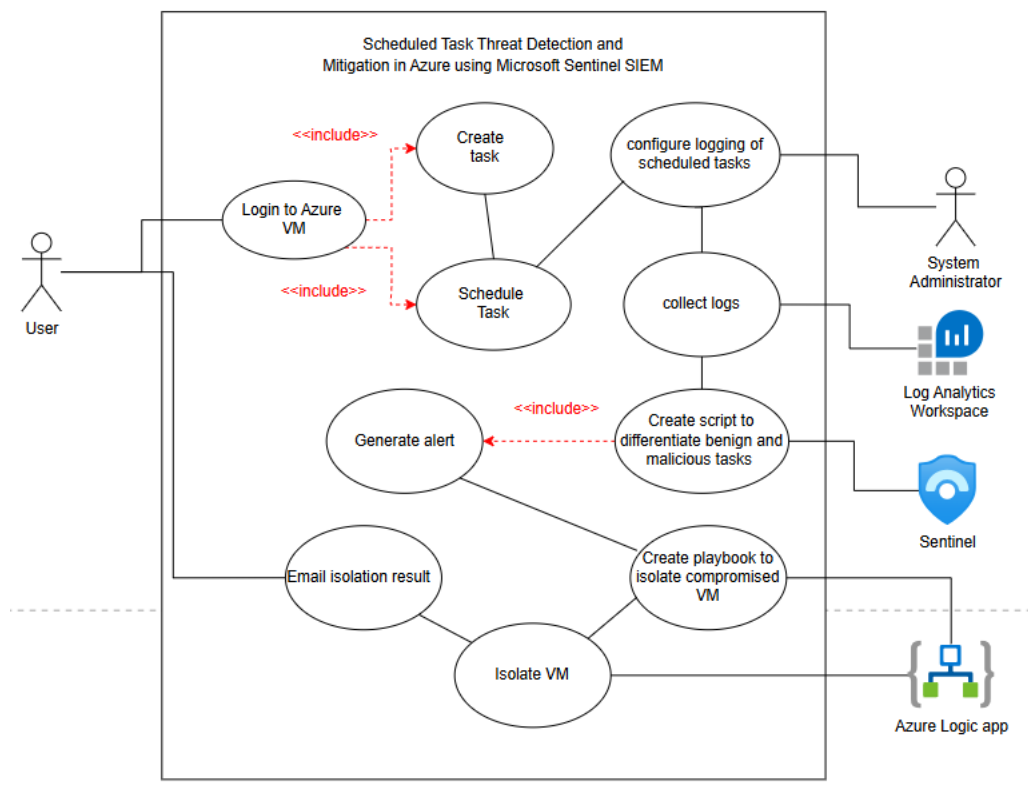


Figure 4.3 Use Case Diagram

4.4.3 Sequence Diagram

Figure 4.4 visualizes the interaction between different components of the system during the detection and mitigation of malicious scheduled tasks. It includes log collection from the VMs, log ingestion, analysis and storage, Microsoft Sentinel uses these logs from the centralized Log Analytics workspace. Data analysis and threat detection occur in Microsoft sentinel using configured rules, alert generation when a threat is detected, and finally automated response implemented by a custom playbook triggered by the alert and finally VM isolation.

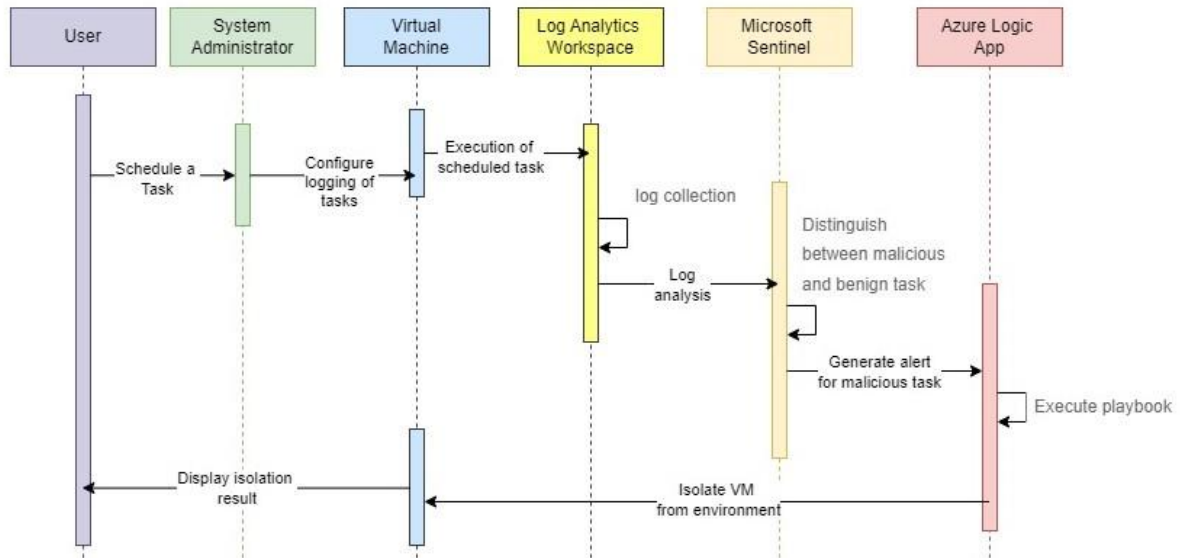


Figure 4.4 Sequence Diagram

4.4.4 Flow Chart

Figure 4.5 provides a view of the data flow within the system. This involves data collection: the flow of log data from VMs to the detection scripts, detection process, the steps taken to analyse the data and identify potential threats, decision points, key decision points where the system determines whether a task is malicious or benign, and finally response actions which are actions taken in response to detected threats, including the isolation of compromised VM.

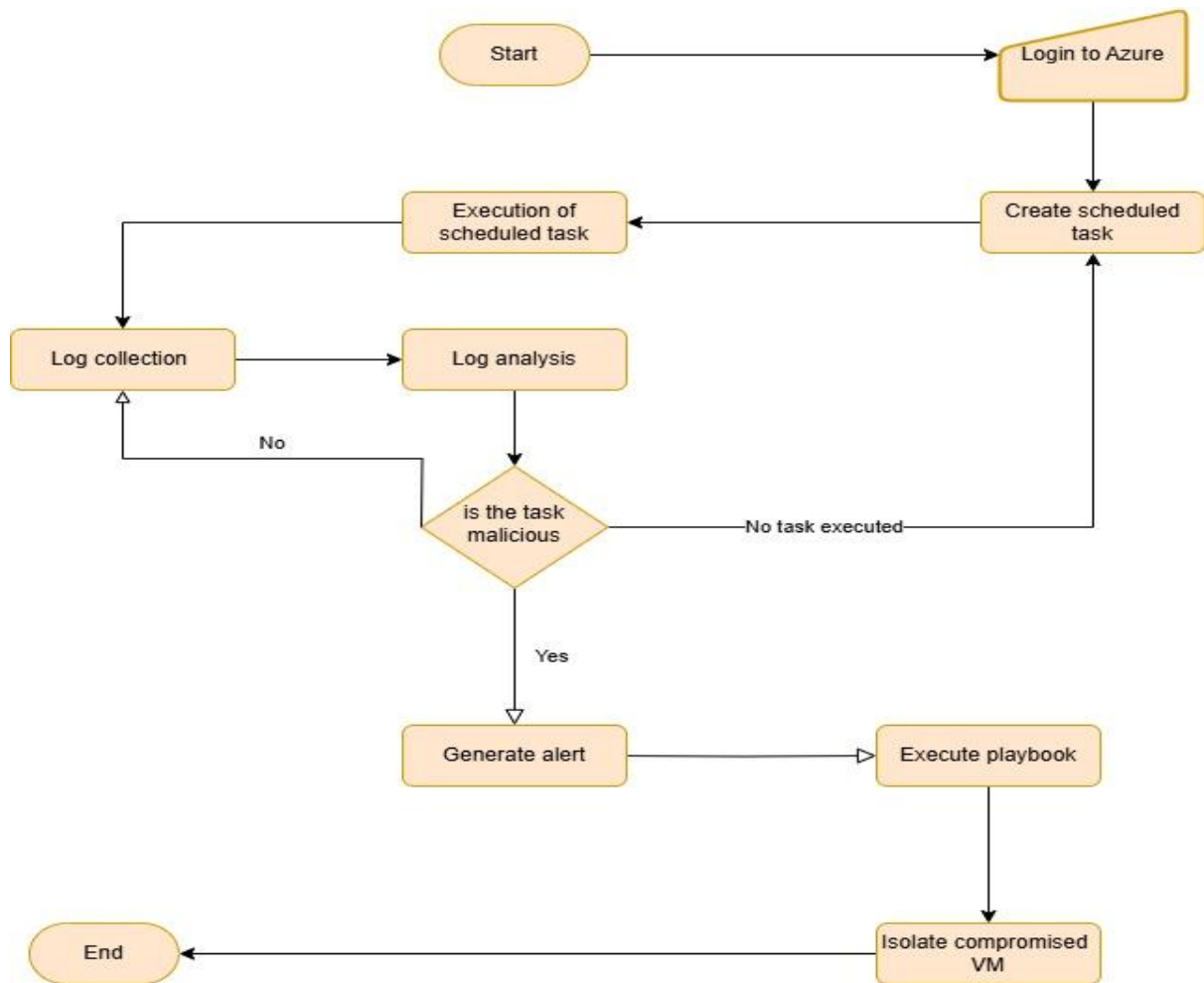


Figure 4.5 Flowchart diagram

4.4.5 State Transition Diagram

Figure 4.6 below describes the different states the SIEM system goes through. From idle to log collection when tasks are executed on the VM and logs are collected. This is followed by analyzing logs to threat detection. After a threat is detected, the state moves from threat detected to alert generation and finally VM isolation.

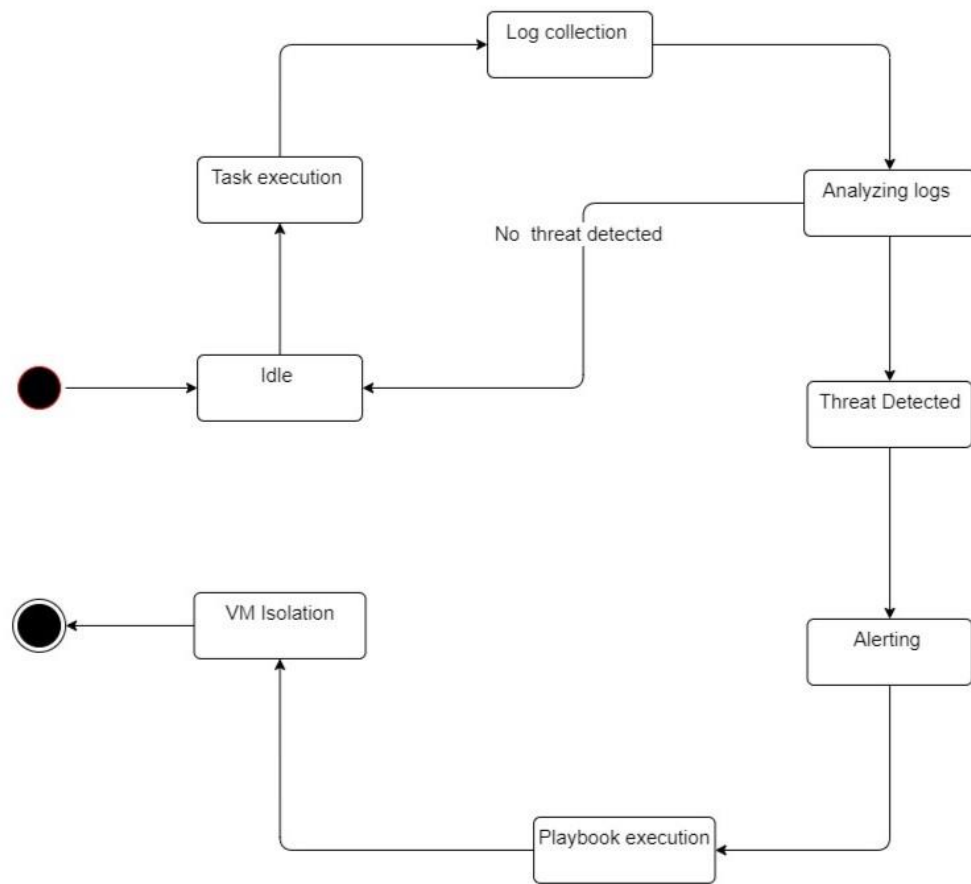


Figure 4.6 State Transition Diagram

Chapter 5: Implementation and Testing

5.1 Introduction

This chapter outlines the implementation and testing of the SIEM system designed to detect and mitigate malicious scheduled tasks on an Azure-based virtual machine. It provides details on the environment used for implementation, including hardware and software specifications, and discusses the testing environment and processes for functional and non-functional requirements. The chapter concludes with an analysis of the results, evaluating the accuracy, efficiency and scalability of the developed system and suggests possible improvements.

5.2 Implementation Environment

5.2.1 System Environment

The SIEM was implemented in a cloud environment specifically Microsoft Azure, using Azure Sentinel and Azure Logic Apps for threat detection and automated response. This environment allowed for secure testing of scheduled task-based attacks and detection mechanisms. The virtualized setup mimicked a corporate infrastructure, with a Windows VM acting as the target system for monitoring and testing purposes. The use of Azure cloud offered scalability and analytic capabilities for testing the detection mechanism developed.

5.2.2 Hardware Specifications

This project makes use of two machines. First, a Lenovo laptop used to RDP (remote desktop connection) to the virtual machine in the Azure network, where the tasks are scheduled, and the second machine was the VM launched in Microsoft Azure Cloud named SentinelVm. The specifications of these machines have been indicated in Table 5.1 below.

Table 5.1 Hardware Specifications

Hardware	Lenovo	SentinelVm
Processor	Intel(R) Core (TM) i5-1035G1 CPU @ 1.00GHz 1.19 GHz	E2-standard, 2 vCPU
Memory	4.00GB	8GB
Storage	5GB	100GB
Network Interface	Intel(R) Wi-Fi 6 AX201 160MHz	gVNIC
Operating System	Windows 10 Pro	Windows 10

5.2.3 Software Specifications

The VM used in this project ran on Microsoft Azure which was chosen due to its compatibility with software used for creating an SIEM and availability of tools that can be configured to automate a response to a scheduled task attack. The software used include:

Log Analytics Workspace (LAW) is a central repository in azure where log data is collected, stored and analysed. It provides a platform for querying and managing log data. LAW was configured to ingest logs from SentinelVm specifically windows security event logs to detect the creation and execution of malicious scheduled tasks.

Sentinel was configured to enable threat detection, investigation and response through analysing collected logs and integrating the SIEM with azure logic apps. Sentinel was used to define and run analytic rules to differentiate between malicious and benign scheduled tasks using detailed metadata and generate alerts that trigger an automated response.

Logic Apps is a cloud-based service used to automate workflows and integrate services and applications Logic apps was used in this project to automate response actions in the event of a malicious scheduled task. A playbook was created that was triggered by sentinel alerts to isolate the compromised VM and inform the network administrator about the isolated VM.

KQL (Kusto Query Language) was used in log analysis to write queries used to parse security event logs to extract relevant details such as task names, file paths, user accounts and associated process IDs. Sentinel analytic rules were also written using KQL allowing customized detection of specific threats eg scheduled tasks with abnormal file paths.

5.3 System Implementation

The implementation of the proposed system involved a step-by-step process to set up the environment, integrate tools, and automate workflows for detecting and mitigating malicious scheduled tasks.

The first step was launching the virtual machine in azure cloud. The machine was assigned a resource group and assigned a virtual network. Rules were configured to the network to allow RDP traffic in the network thus allowing the Lenovo laptop to connect to the virtual machine. Two scripts were created to execute on the VM. One was a benign backup script set by a system administrator and the other was a malicious script. named the same as the backup script, used to steal data from the virtual machine. These two scripts were named in a way that could easily be missed by an administrator but not the SIEM thus increasing the accuracy of the system and reducing false positives.

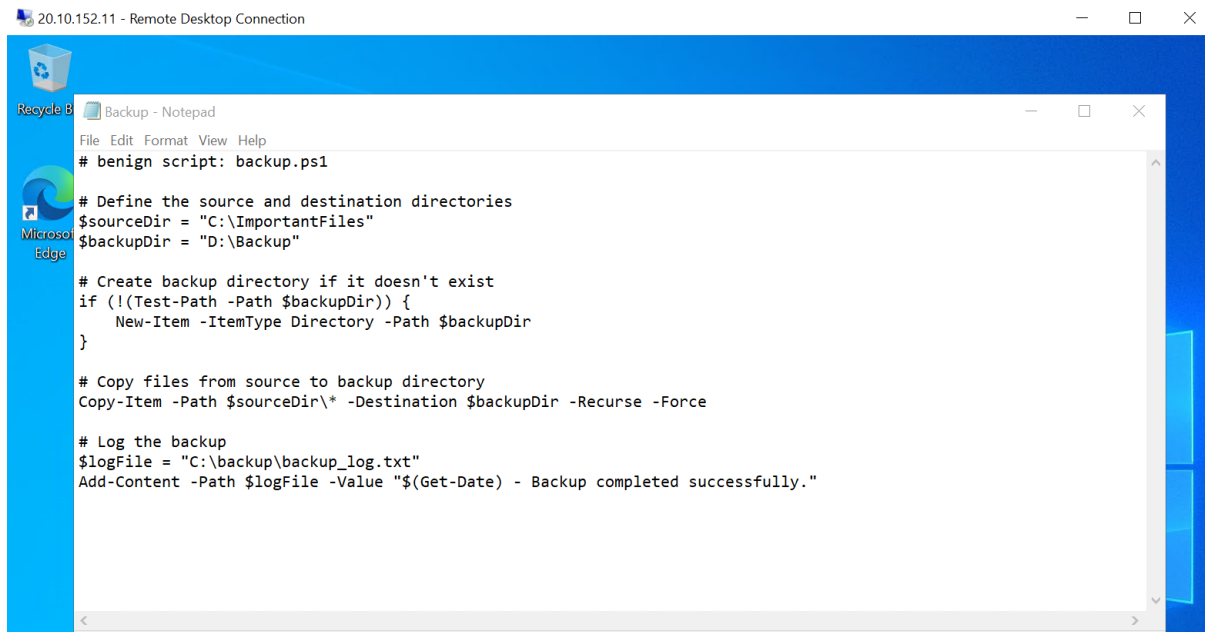


Figure 5.1 Benign script

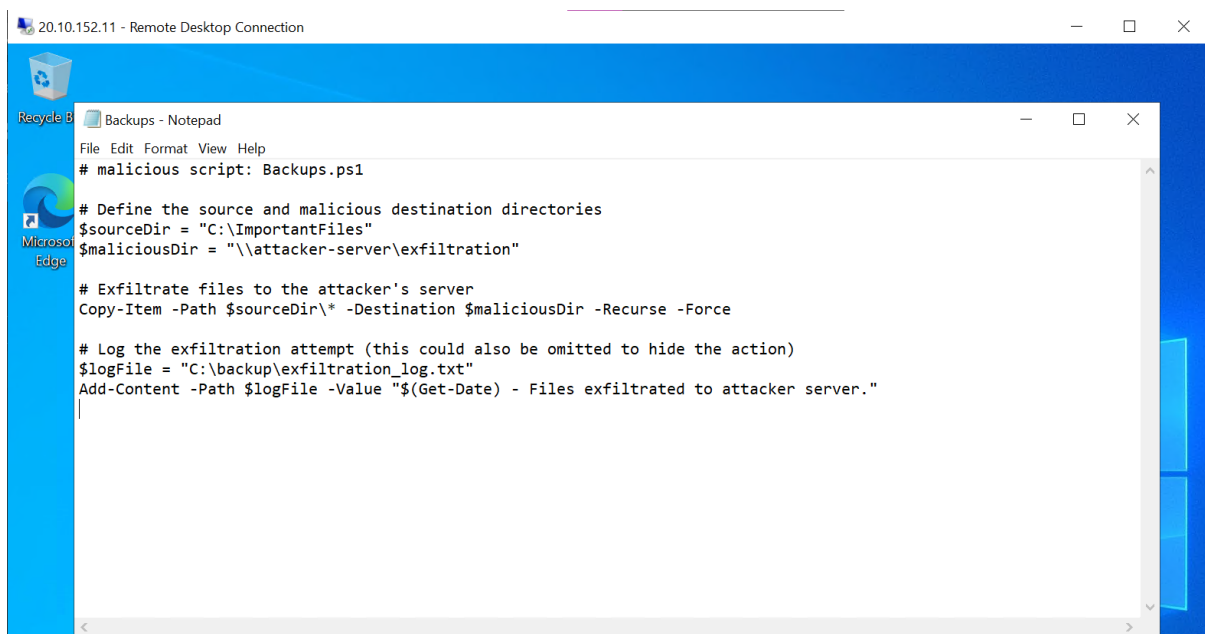


Figure 5.2 Malicious Script

The scripts were scheduled to execute at a specific time close to each other in order to test whether the SIEM would differentiate them and trigger an alert based on the execution of the correct (malicious) script.

The next step was setting up log analytics workspace, configuring it with a unique name and connecting it to the VM to enable data collection of windows security events. The workspace

was configured to ingest all security event logs collected from the virtual machine, and using specific KQL queries you could get any information on the tasks scheduled.

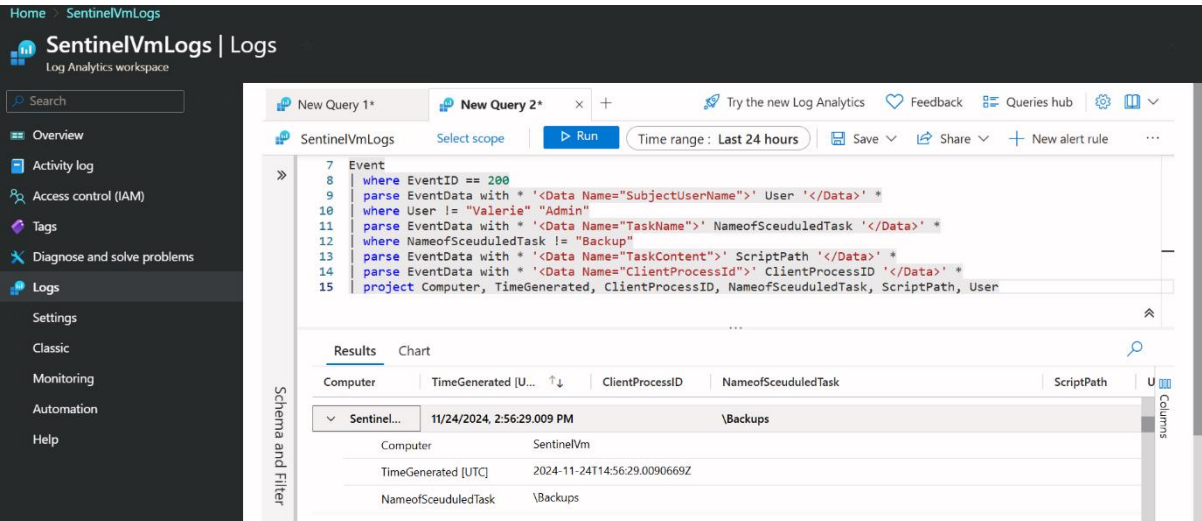


Figure 5.3 Log Analytics Workspace

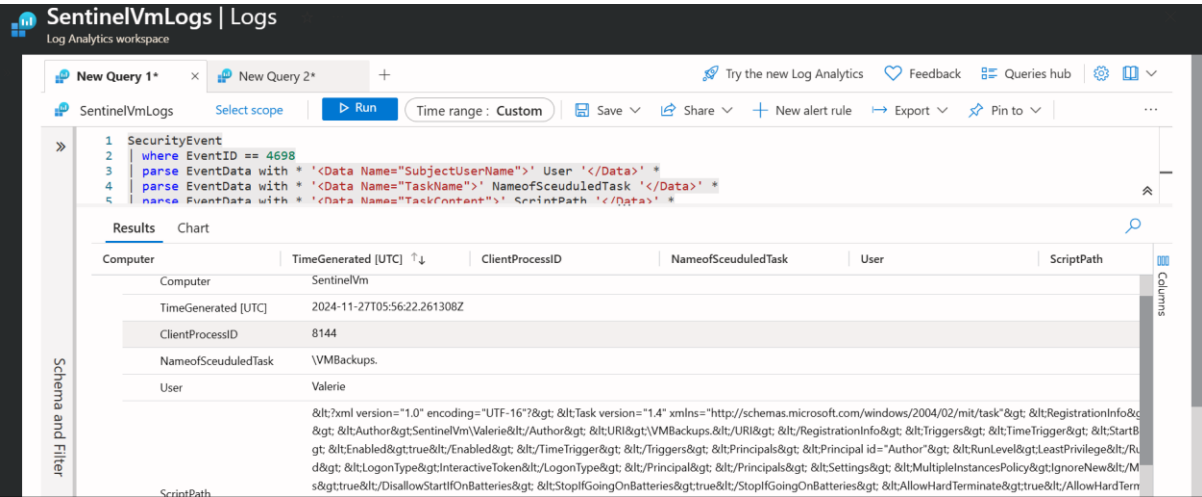


Figure 5.4 Query Results

Once the workplace was set up, Sentinel was connected to the workspace and custom analytic rules were created using KQL in Sentinel to detect suspicious scheduled tasks activities and filter log data for specific details such as task names, user accounts, and script paths. The rules were colour coded according to their severity, red indicated high severity meaning a malicious task was executed and yellow was medium severity meaning a task was created and executed in the vm but was not malicious. Conditions were defined in the rules in sentinel to flag abnormal behaviour such as unidentified scripts that are executed (Backups) and if detected an alert configured in sentinel is triggered.

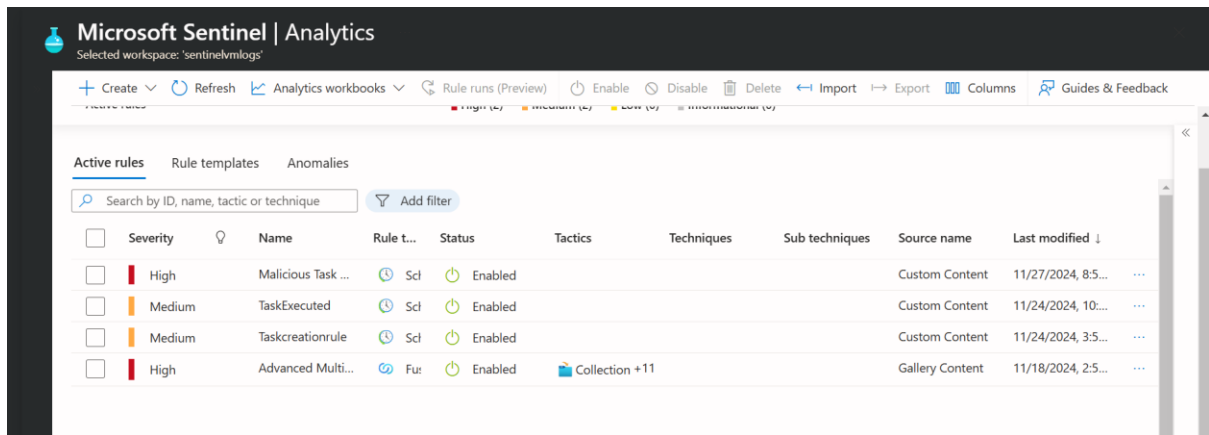


Figure 5.5 Analytic Rules

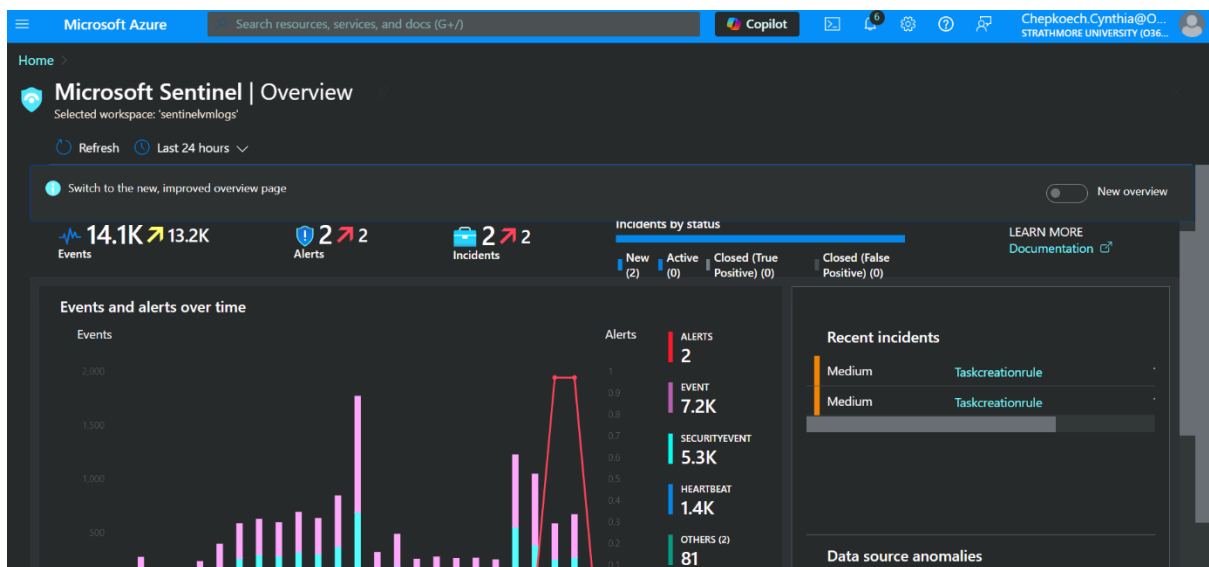


Figure 5.6 Sentinel Alerts View

Next a Logic App named VMIsolation was created to automate the systems response to detected threats. A playbook was created in the app that was configured with the trigger “When an alert is triggered in Sentinel” and workflow steps were defined to isolate the VM from the network by deallocating it and shutting it down then a notification with details of the incident is sent to the administrators.

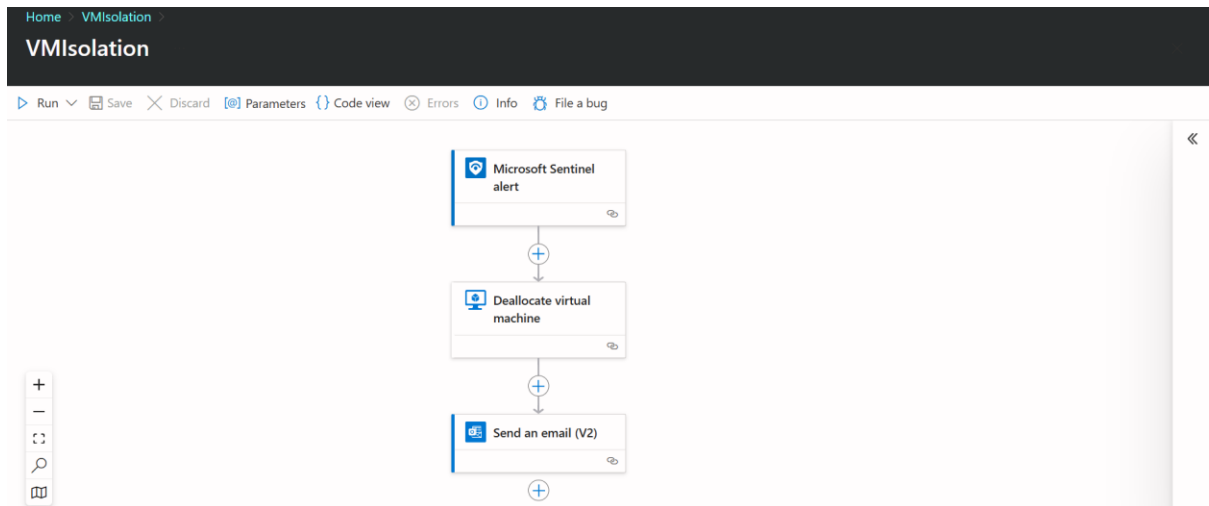


Figure 5.7 Playbook Workflow

5.4 Testing Environment

5.4.1 Testing of Functional Requirements

Table 5.2 provides a summary of the testing conducted to evaluate the functional requirements of the system. Each test case examines a specific function defined in Chapter 4, verifying the systems capability for detecting and mitigating scheduled task-based attacks.

Table 5.2 Functional Requirements Testing

Test Case	Description	Test Data	Results	Test Verdict
1	The system must log all detected events related to scheduled tasks.	Executed scheduled tasks, both benign and malicious	LAW captured logs from the VM and included metadata such as execution details	Pass
2	The system should identify malicious tasks by analysing the logs and differentiate them from benign tasks	A malicious script simulating system backup was scheduled to run alongside the real backup script	The malicious script is flagged as suspicious while the benign task is ignored.	Pass
3	Upon detecting a malicious scheduled task, the system should isolate the	Alert generated in sentinel after a malicious task is detected	The Logic app configured successfully isolates the VM by	Pass

	compromised VM automatically		deallocating it and shutting it down.	
4	The virtual machine and logic apps should integrate seamlessly with sentinel for monitoring and threat response	Sentinel monitors logs from the VM and triggers alerts to be used by the logic apps	Alerts are generated in sentinel once a malicious task is detected from the logs and the alert triggers the isolation of the VM by the logic app.	Pass

5.4.2 Testing of Non-Functional Requirements

Table 5.2 highlights the testing of non-functional requirements, including performance, accuracy, and reliability of the system.

Table 5.3 Non-Functional Requirements Testing

Test Case	Description	Test Data	Results	Test Verdict
1	The system should be accurate in detecting malicious scheduled tasks	A mix of malicious and benign backup tasks were tested	System correctly identified malicious tasks using logs metadata	Pass
2	The performance of the system should be high providing real-time monitoring and analysis	A simulated malicious and benign scheduled task are simulated	Real-time alerts were generated in sentinel with minimal latency	Pass
3	The system should be reliable enabling it to handle large volumes of data	All windows security events are collected from the VM and categorised in LAW	The system successfully categorises scheduled tasks events and detects suspicious tasks	Pass
4	The system must ensure continuous	All tasks executed on the virtual machine	The system successfully logs all	Pass

	monitoring for immediate threat detection		tasks executed on the virtual machine	
--	---	--	---------------------------------------	--

5.5 Discussion of Results

The outcome of the functional and non-functional testing show that the SIEM solution was successful in accurately detecting and responding to scheduled task-based threats in a cloud environment.

According to the functional requirements, the system was able to collect all security event logs from the virtual machine. Specific KQL queries enabled categorization of the logs to find which tasks created by a specific user and more details or metadata about the tasks as seen below.

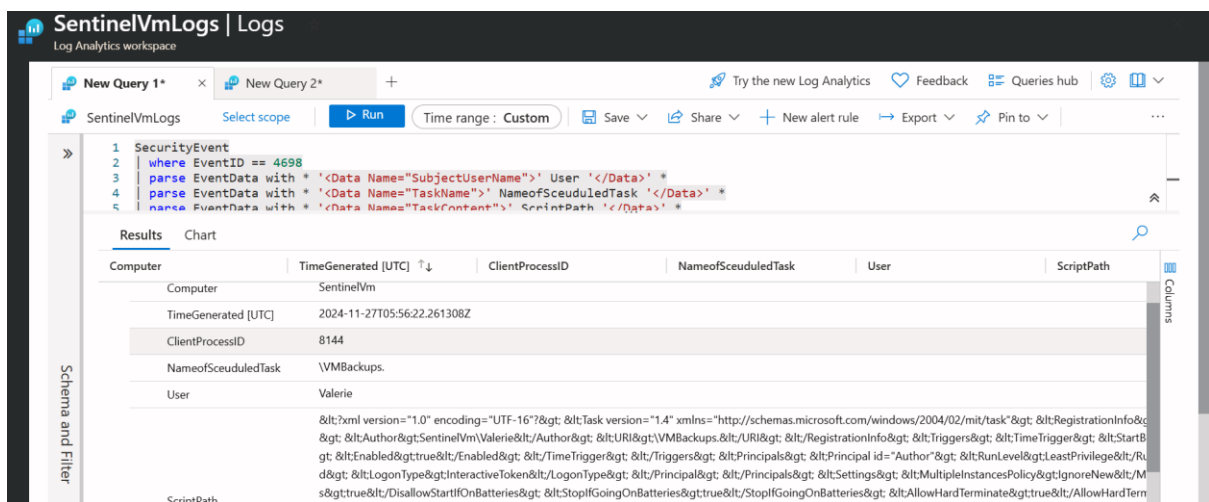


Figure 5.8 Log Metadata

The system distinguished between a benign and malicious task reliably. When a malicious task is executed, the alert set by the analytic rule discussed in system implementation is triggered as seen below (red). The alert is connected to log analytics and more information about the task executed to provided as seen in appendix 2 and 3.

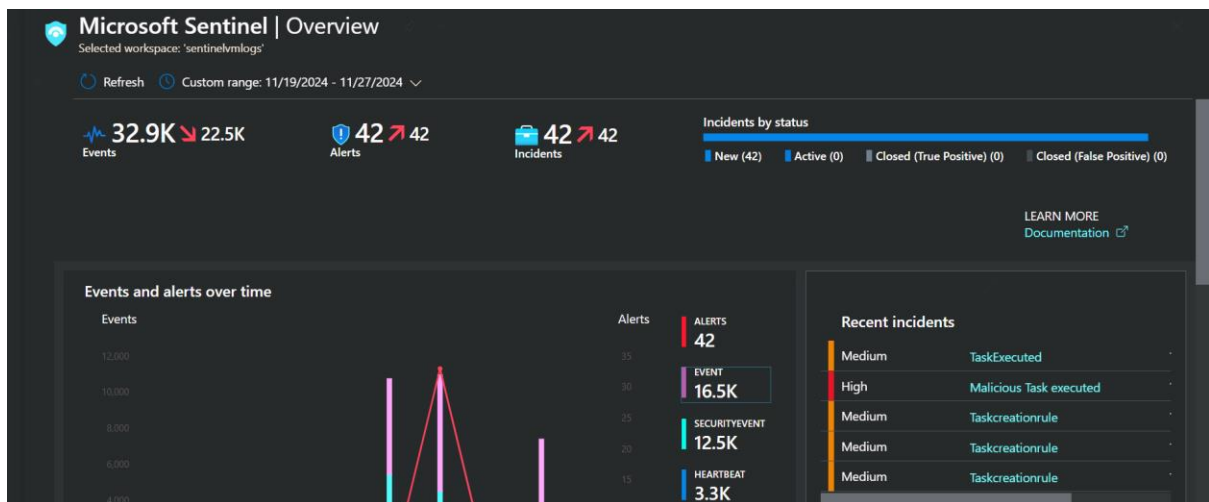


Figure 5.9 Alerts on Sentinel

The system was able to detect the malicious task executed and the alerts created triggered automated response.

General Set rule logic Incident settings **Automated response** Review + create

Automation rules
View all automation rules that will be triggered by this analytics rule and create new automation rules.

+ Add new

Order	Automation rule name	Trigger	Action	Status	
1	VmIsolation	Alert created	Run playbook 'VmIsolation'	Enabled	...

Figure 5.10 Automation Configuration

The detection of the malicious task triggered the automated response (playbook) using azure logic apps which isolated the compromised VM in real time within minutes of detecting the malicious task executed in the virtual machine, thus fulfilling its nonfunctional requirements. This can be seen below in the time when the alert was generated, and the email was sent to the administrator.

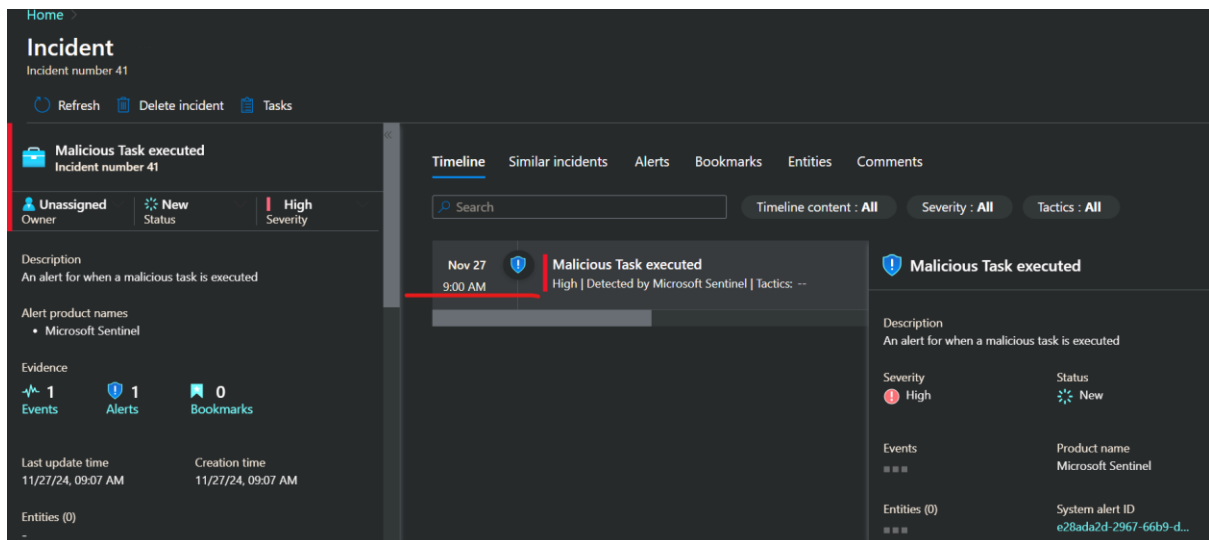


Figure 5.11 Alert Generation Time

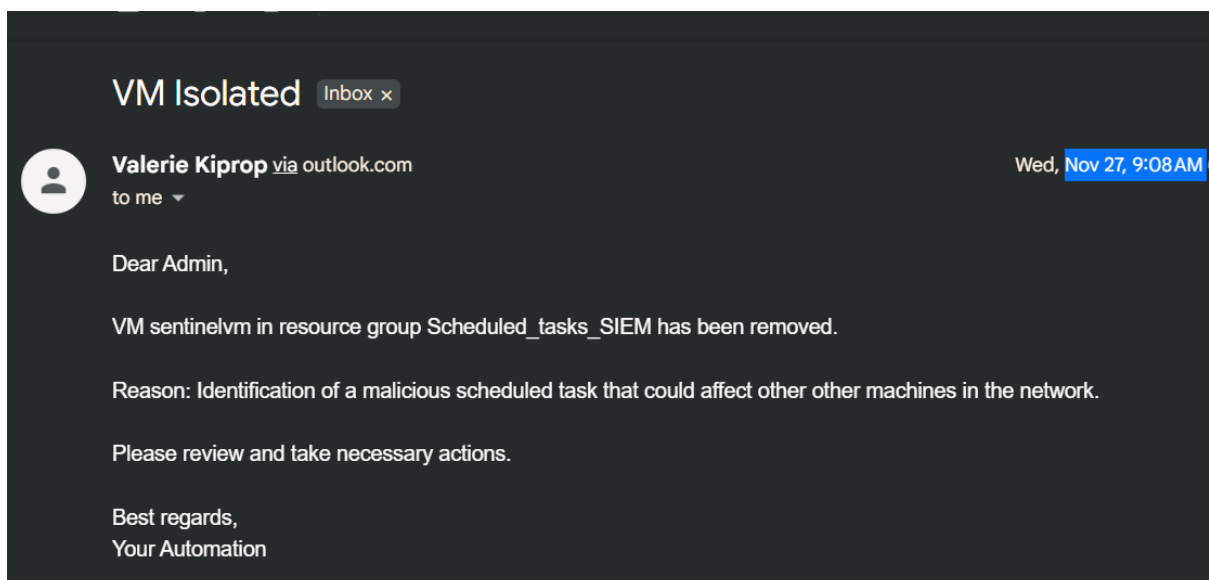


Figure 5.12 Email to Administrator

In conclusion the SIEM was able to detect the malicious scheduled task in the virtual machine and mitigate it by isolating the virtual machine to avoid further compromise to other machines in the cloud network.

Chapter 6: Conclusions and Recommendations

6.1 Conclusion

This project has highlighted the need for proactive measures in addressing security threats posed by malicious scheduled tasks in the cloud environments. By utilizing Sentinel, Logic apps and Log analytics workspace, the system demonstrated the ability to provide real-time detection and automated mitigation of such threats. The use of tools such as KQL enabled querying and analysis of logs, thus improving threat detection and accuracy in minimizing false positives.

6.2 Recommendations

While the project performed as per the expectations, there are a few recommendations to further enhance its applicability to the real world. First the test environment of the system can be improved by introducing multi-tier networks and diverse endpoint configurations to mirror real-world enterprise setups.

Machine learning models can also be integrated into the detection process to identify emerging attack patterns and adopt to new threats in real-time.

Lastly the systems detection logic can be improved by incorporating additional data sources such as network traffic logs for a more extensive analysis.

6.3 Future Works

This project has laid a foundation for scheduled task-based threat detection but has also opened avenues for future research and development. Some potential areas of exploration include a broader scope of persistent threats, expanding the focus beyond scheduled tasks to other forms of persistence mechanisms such as registry changes, startup programs and service modifications. The systems capability can also be expanded to non-windows environments such as Linux or MacOS to detect similar persistence mechanisms on diverse operating systems.

By pursuing these recommendations and future work opportunities, this project can evolve into a versatile security solution that addresses a wide range of cybersecurity challenges.

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Appendices

Appendix 1 Logs

Home > SentinelVmLogs

SentinelVmLogs | Logs

Log Analytics workspace

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Logs**
- Settings
- Classic
- Monitoring
- Automation
- Help

New Query 1* New Query 2* x + Try the new Log Analytics Feedback Queries hub

SentinelVmLogs Select scope Run Time range: Last 24 hours Save Share + New alert rule

```
7 Event
8 where EventID == 200
9 parse EventData with * '<Data Name="SubjectUserName">' User '</Data>' *
10 where User != "Valerie" "Admin"
11 parse EventData with * '<Data Name="TaskName">' NameofSceuduledTask '</Data>' *
12 where NameofSceuduledTask != "Backup"
13 parse EventData with * '<Data Name="TaskContent">' ScriptPath '</Data>' *
14 parse EventData with * '<Data Name="ClientProcessID">' ClientProcessID '</Data>' *
15 project Computer, TimeGenerated, ClientProcessID, NameofSceuduledTask, ScriptPath, User
```

Results Chart

Computer	TimeGenerated [UTC]	ClientProcessID	NameofSceuduledTask	ScriptPath	User
Sentinel...	11/24/2024, 2:56:29.009 PM		\Backups		
Computer	SentinelVm				
TimeGenerated [UTC]	2024-11-24T14:56:29.0090669Z				
NameofSceuduledTask	\Backups				

Schema and Filter

Appendix 2 Task Executed Information Logs

Home > Microsoft Sentinel | Overview > Incident >

Logs

SentinelVmLogs

New Query 1* x + Try the new Log Analytics Feedback Queries hub

SentinelVmLogs Run Time range: Custom Save Share + New alert rule Export Pin to Format query

```
1 // might contain sensitive data
2 let alertedEvent = datatable(compressedRec: string)
3 [
4   'eAGVvtu2zAM/ZXAD31aY+fwtC6Gom261kDaBLm0wOAXzWYcIb3kSHLSYti/j7ITX3LD9mieQ5o8pcj9th5FnkQapOVaU+CacmDvsfXNmtEYnoGDJBpCBNT0u3vZa122+zOn5/
   au3Ha72b5qdZxr5yFShx1F77EUASj1DdDhutXtIvBGYhCLaQ8pmdIIZ0StEPT999cHeqzSRDWRNA0kTF5Y6CViF0zf3n3GrLEGqajg332r1XR8qwE8ECH1ERmsx
   +XrSvFuruI9G3DeJjANY8uemAUrvBrqXXi2rYKlIhAT1YxpIIUC90MRGxvKA/Frtltx+naTtu0QbY1RvOtLHgJcz+B1CqNymB4jy/EFsrBAWpkDFWnr1e1+11mp3+Tb/
   ...
5 ]
```

Results Chart Add bookmark

TimeGenerated [UTC]	ClientProcessID	Computer	NameofSceuduledTask	ScriptPath	User
11/27/2024, 5:56:22.261 AM	8144	SentinelVm	\VMBackups.	<?xml version="1.0" encoding="UTF-16"?><Task version="1.4" xmlns="...	Valerie
ClientProcessID	8144				
Computer	SentinelVm				
NameofSceuduledTask	\VMBackups.				

Schema and Filter

Appendix 3 Malicious Task Executed Information Logs

Schema and Filter

Logs

SentinelVmlLogs

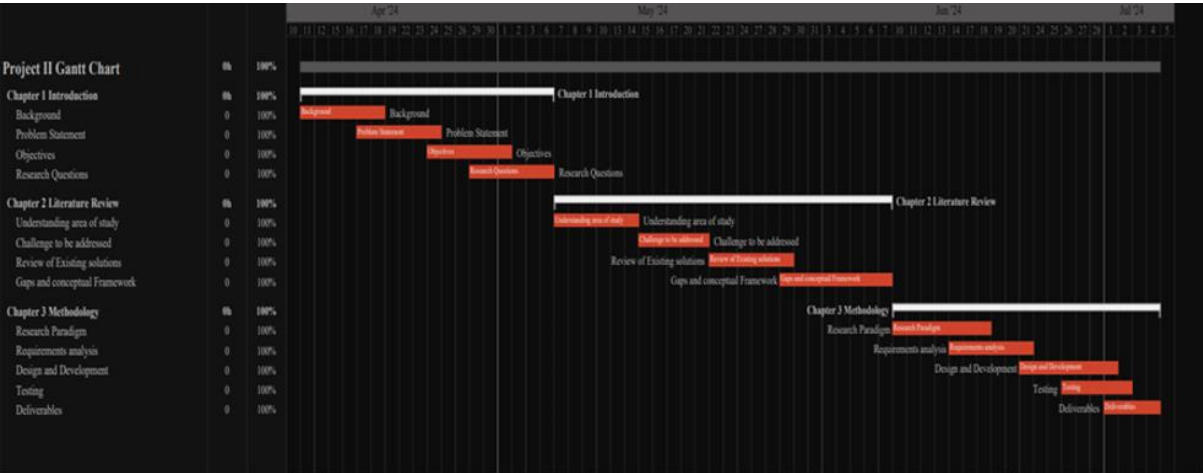
Results

Chart

Add bookmark

☐	TimeGenerated [UTC] ↑↓	_ResourceId	Computer	EventCategory	EventData
		EventData			<DataItem Type="System.XmlData" time="2024-11-27T06:00:01.0119821Z" sourceHealthServiceId="51d13401-a1fa-4e7e-bb4b-bbeb62865125"><EventDataItem>
		EventID			200
		EventLevel			4
		EventLevelName			Information
		EventLog			Microsoft-Windows-TaskScheduler/Operational
		ManagementGroupName			AOI-9ae804ec-8f9e-4399-9016-6bb618bf5884
		MG			00000000-0000-0000-0000-000000000001
		ParameterXml			<Param><VMBackups></Param><Param>C:\Windows\System32\notepad.exe</Param><Param>[c1997679-1bf2-48f7-9259-b16564c9c819]</Param><Param>
		RenderedDescription			Task Scheduler launched action "C:\Windows\System32\notepad.exe" in instance "[c1997679-1bf2-48f7-9259-b16564c9c819]" of task "VMBackups."

Appendix 4 Gantt Chart



**BSC. IN COMPUTER NETWORKS AND CYBER SECURITY &
TELECOMMUNICATIONS**

FINAL REPORT MARKING GUIDE

CNS 4101: Cyber Security and Computer Networks Project II

Section A – Student’s Details:

Student Number	Student Name
146640	Kiprop, Valerie Cheruto
Project Title:	

Section B – For examiner’s use only:

Area of Assessment	Score	Total
Abstract (3)		
- Describes the concept clearly, including the outcome of the research (3)		
Introduction (7)		
- The problem was clearly identified and it relates to a Computer Network and Cyber Security challenge? (2) - Are the objectives S.M.A.R.T and incremental? (2) - Do the objectives support the achievement of the project’s aim? (1) - Has the significance of the project and justification for undertaking the project been clearly explained? (1)		

- Scope proves candidate has learnt significantly from doing the project (1)		
Literature Review (8)		
- Has the objective-literature mapping (Literature Review sections to address each Specific Objective) been done correctly? (3) - Does the literature review analyse what has been done before in relation to the project, what has not been done (the gap), and what the project proposes to do to address the gap? (3) - Has adequate literature been reviewed? (2)		
Methodology and System Analysis & Design (9)		
- Has the correct/fitting design methodology been used. (2) - Has the initial system design supported with a valid design diagram been presented? (2) - Have the functional and non-functional requirements, system architecture, and System Analysis Diagrams been discussed? (5)		
Implementation (6)		
- Presents practical implementation with explanation of actual equipment, software etc used to implement the work based on the design in chapter 3&4 (6)		
Testing and Results (6)		
- The system/model/etc was tested and all test results have been presented (3) - Observation and analysis of the results have been discussed in detail. (3)		
Overall Assessment (11)		
- Is there a logical mapping from the Title to the Objectives, to the Literature Review topics, to the Methodology, Implementation, and results obtained? (4) - Proposal document is neat and well formatted. Table of contents numbered in sequence and corresponds to page Numbers (2) - Evidence that figures were drawn by candidate and not copy pasted (2) - Has the APA referencing style been correctly applied? (1)		

- Demonstrates that the candidate has learnt from the project (2)		
TOTAL (50)		

Remarks:

Examiner (initials)

Date:
